

Project Proposal Guidelines

The project proposal is the roadmap for a project worth 30% of your total grade.

- Successful proposals will lead to a successful final report.
- Ensure the document is professional as presentation quality (organization and clarity) is part of the grading criteria!.

Ideas/Examples

- Smart City: Digital Twin for Urban Heat Mitigation
 - Identify "heat islands" in a specific city and propose optimal locations for new green infrastructure
- Demand-Driven Logistics: Optimizing "Dark Store" Locations
 - Dark stores are micro-fulfillment centers, minimize delivery times during peak business hours.
- Disaster Response: Flood Risk for Critical Infrastructure
 - Evaluate the vulnerability of emergency services (hospitals, fire stations) during a 100-year flood event.
- *Mooglegaps* <https://mooglegaps.netlify.app/>
- *Radio Garden* <https://radio.garden/>

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Purpose

- The project proposal is the first formal step in your applied spatial analysis project.
- In groups of 2, you will design a solution to a practical engineering problem using spatial data engineering techniques.
- The proposal "ensures" your team has a viable plan for data integration, analysis, and visualization.
- Concise document (submitted on Canvas) covering the following parts:
 - Problem Statement
 - Data Sources & Integration
 - Methodology
 - Analysis Plan
 - Visualization Goals

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Required Components

- **Problem Statement:**
 - Define a practical engineering problem that can be addressed through spatial analysis.
- **Data Sources & Integration:**
 - Identify the "large-scale" spatial datasets you plan to use.
 - Describe how you will perform Spatial ETL (Extract, Transform, Load), including handling coordinate reference systems (CRS).
- **Methodology:**
 - Outline the spatial and attribute operations required.
 - Explain how you will model data structures to reflect spatial relationships.
 - If applicable, describe plans for Raster-Vector interaction or Data Cube construction.
- **Analysis Plan:**
 - Describe the specific spatial statistics or modeling techniques (e.g., interpolation, spatial regression, or suitability analysis) you intend to use.
- **Visualization Goals:**
 - Briefly describe how you plan to communicate results (e.g., report with high-resolution maps or a Story Map).

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Expectations & Standards

- **Reproducibility:**

- Your final project must include an automated workflow and reproducible code.
- Use the proposal to identify the tools (e.g., R, Python, or Docker) and libraries you will use to ensure this.

- **Ethics & Privacy:**

- You must briefly note any licensing or ethical considerations related to your chosen datasets.

- **Academic Integrity:**

- All work must reflect your team's independent effort.
- While AI can be used for brainstorming or clarifying your writing, it cannot be used to generate the substantive analysis or solve the broad programming tasks of the project.